Iteration 1 Report

Meal Planner App for the Elderly (MealCare)

Programme: B.Sc. in Software Development

Module: Third Year Software Development Project

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Submission Date: 26 February 2026

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1. Iteration Objectives

In the first iteration, the main focus was to build the foundation of the system. The goals were to create a working **user profile system**, a **nutrition based filtering engine**, and a **simple meal planning UI** using React. The idea was to allow users to select their dietary preferences, health conditions, and budgets, and then generate meal options that match those details. This step also helped set up a clean and scalable architecture for the next stages of development.

2. Features Implemented

2.1. Profile Management System

A profile form was developed where users can select their age group, dietary preferences like vegetarian or plant based, allergies like nuts, dairy, and health conditions such as high blood pressure or diabetes. They can also enter a budget preference.
From a technical side, controlled components and context handling were used to manage profile data efficiently. Data is stored locally in the browser so it stays available when users reload the app.

2.2. Application Architecture

To make the project more organized, a layered structure was designed:

- **Context Layer** provides global profile data without mixing it with UI logic.
- **Service Layer** includes all the filtering logic such as dietary, allergens, budget, etc.. written separately from React code.
- **Hook Layer** connects the UI with the filtering logic using custom hooks.
- **Page Layer** manages app navigation such as Home, Meal Plan, Profile, Shopping.

This structure ensures the app is easy to scale and maintain as new features are added.

2.3. Nutrition Filtering Engine

This engine filters meals dynamically based on the user's profile. For example, it removes meals containing allergens, avoids high sodium or sugary meals for users with certain health conditions, and considers budget preferences. The filtering updates the interface in real time whenever profile data changes.

2.4. Meal Categorization and Display

Filtered meals are grouped under **Breakfast, Lunch,** and **Dinner** categories. Each section shows the number of meals available and displays a **No meals found** message if none match.

2.5. Weekly Meal Planning System

Users can plan meals for every day of the week **Monday to Sunday**. The system uses two state levels:

- A temporary editing state for current selections, and
- A saved state for each day's confirmed plan.
  The **Save for Day** button only allows saving when all meal categories are selected, ensuring complete daily plans. The system also remembers meals when switching between days without mixing data.

2.6. User Interface Enhancements

A reusable **MealCard** component was built to display meals with a clear **Select** or **Selected** button. A **progress bar** shows overall planning completion, and the layout adapts well to different screen sizes.

3. React and JavaScript Concepts Used

Throughout the iteration, several concepts were applied:

- **React Hooks:** useState, useEffect, useContext, and custom hooks for data handling.
- **State Management:** Nested and functional state updates were used for complex data like weekly plans.
- **JavaScript Functions:** Methods like .filter(), .map(), and .some() were used to manage lists and apply business logic.

4. Changes to Vision Document

The Vision remains unchanged, but implementation priority was adjusted to establish an intelligent meal planning system for elderly users remained the same. However, this iteration shifted focus toward building a profile based filtering system first.

5. Changes in Research and Design (R&D)

Some small decisions were made during this stage:

- Adopted a **layered architecture** (Context → Hook → Service).
- Chose to use a local JSON file instead of a live API during testing.
- Deferred integration with Supabase (database) to a later iteration.

6. Changes in Specifications

The specification remains were unchanged but certain requirements were refined during implementation to clarify system behaviour and improve reliability.

- Added meal saving logic per day.
- Validation before saving plans.
- Count display for meal categories.

7. What Was Not Completed

Some features were postponed to the next iteration, such as:

- Shopping list creation.
- Integration with Supabase.
- Viewing and editing saved plans.
- User authentication system.

8. Iteration Summary

Iteration 1 established the core structure of **MealCare**. The profile management system, nutrition filtering logic, and weekly meal planner are fully functional. The system now provides a strong, scalable foundation ready for **Iteration 2**, which will introduce automatic plan creation and more advanced features.

9. Revised Iteration 2 Plan

- The second iteration will focus on extending the system with shopping and data persistence features. The primary priority is implementing smart shopping list generation based on confirmed weekly meal plans, including ingredient consolidation and automatic updates.
- Supabase integration will be introduced to replace local storage and enable persistent cloud-based data management for user profiles and meal plans. This will improve reliability and scalability.
- Barcode scanning functionality will be implemented to support shopping list verification during grocery tasks. If time permits, basic user authentication (sign-up, sign-in, login) will be added to enable secure user-based access to stored data.

10. Implemented Figures

The following figures present screenshots of the implemented application at the end of Iteration 1. These images demonstrate the core functional features delivered in this iteration and confirm that a potentially shippable product has been deployed.

The **Fig.1** screen represents the main entry point of the application. It provides navigation to core features such as Profile Setup, Meal Planning, and Shopping (future feature placeholders), demonstrating overall system structure and accessibility-focused layout.

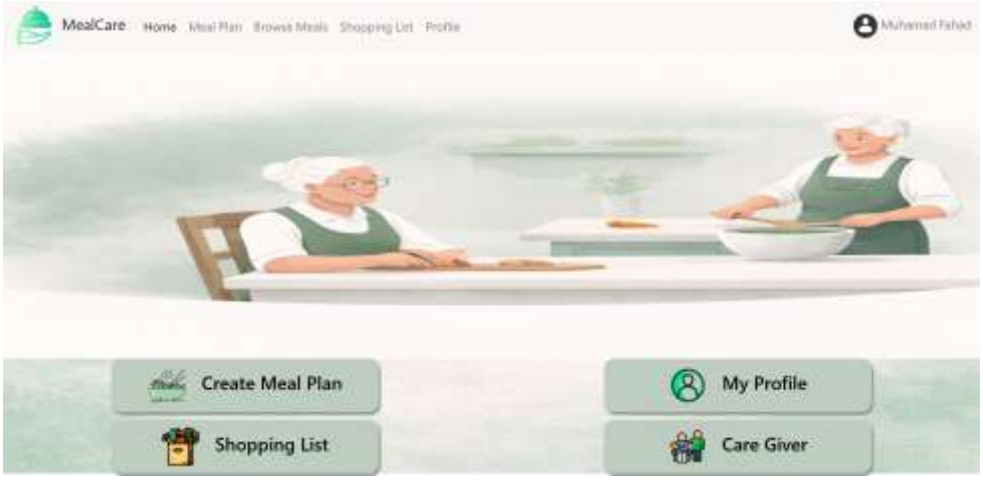

Fig.1: Home / Dashboard Screen (Implemented Version)

The **Fig.2** demonstrates profile creation and preference selection, including dietary requirements and health conditions. The form uses controlled inputs and stores user data for filtering and meal recommendation logic.

Fig. 2: User Profile Management Screen (Implemented Version)

The **Fig.3** and **Fig.4** screen demonstrates the structured weekly planning interface. It shows filtered meal options, daily selection behaviour, validation rules, and progress feedback mechanisms implemented during Iteration 1.

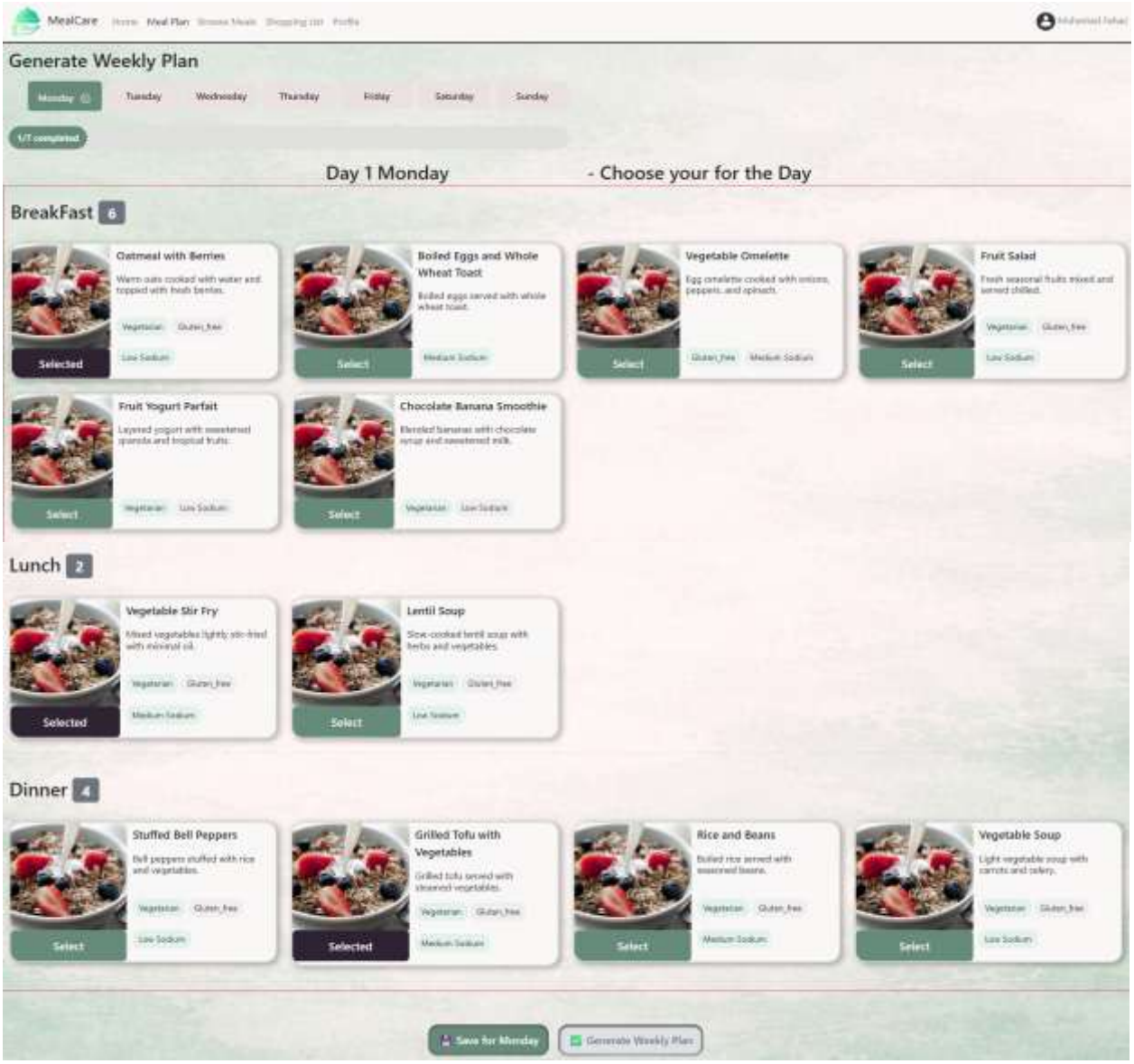

Fig.3: Weekly Meal Planning Screen (Implemented Version)

11. Use of Generative AI

Generative AI tools were used only to assist with content structuring, idea clarification, and summarizing technical details not for direct code generation. Gen AI refers to systems that can produce text, images, or other forms of content based on learned data patterns. In this project, it supported clarity and helped with documentation preparation rather than programming logic.

DECLARATION

*I declare that all material in this submission, e.g. thesis/essay/project/assignment, is entirely my/our own work except where duly acknowledged.

*I have cited the sources of all quotations, paraphrases, summaries of information, tables, diagrams or other material, including software and other electronic media in which intellectual property rights may reside.

***I have cited clearly, exhaustively and adequately in my source code, code generated by Gen AI tools.**

*I understand that I can only get credit for work that I have done (e.g. understanding other code, interfacing my code with it, modifying it, etc.), but you will not be credited with the third party code itself.

*I have provided a complete bibliography of all works and sources used in the preparation of this submission.

*I understand that failure to comply with the Institute's regulations governing plagiarism constitutes a serious offence.

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